full-mhs-z: magnetohydrostatic equilibrium with pressure and field adjusted

The physics of `run.mode = "full-mhs-z"`, explained step by step (B.Sc. level). HSEMHS.

1. The question

We are given a snapshot of the solar atmosphere: the temperature T and the magnetic field $\mathbf{B}_{\mathrm{obs}}$ at every point of a box. In the real Sun (or in the simulation the snapshot comes from) the gas moves, so the forces on it do not cancel: the net force is what accelerates it. We want a **static** atmosphere, with zero net force everywhere, that looks as much as possible like the snapshot: the temperature exactly as given, the magnetic field as close as possible to $\mathbf{B}_{\mathrm{obs}}$, and the pressure (hence the density) adjusted. Such a state is called **magnetohydrostatic (MHS)** equilibrium. Unlike the `mhs-poisson` modes, which change only the pressure, this mode also changes the field. That is what makes a true balance possible, as Sect. 4 explains.

2. The quantities

Symbol	Meaning	Unit
x, y, z	horizontal coordinates (periodic box) and height	m
T	temperature, given, **never changed**	K
P, u	gas pressure (unknown); $u = \ln P$	Pa, –
ρ	density, $\rho(T, P)$ from the equation of state	$\mathrm{kg\,m^{-3}}$
$\mathbf{B}$, $\mathbf{B}_{\mathrm{obs}}$	the model magnetic field (unknown) and the given field	T
$\mathbf{A}$	magnetic vector potential, $\mathbf{B} = \nabla \times \mathbf{A}$ (the actual field unknown)	T m
$\bar{b}$	mean vertical field through the box (one number)	T
$\mathbf{J}$	current density, $\nabla \times \mathbf{B}/\mu_0$	$\mathrm{A\,m^{-2}}$
σ	tolerated deviation of $\mathbf{B}$ from $\mathbf{B}_{\mathrm{obs}}$ (default: $5\,\mathrm{G} = 5 \times 10^{-4}\,\mathrm{T}$ per component)	T
$\mathbf{R}$	net force per volume (force residual)	$\mathrm{N\,m^{-3}}$
S	local weight of the gas, $\approx \rho g$ from the starting state	$\mathrm{N\,m^{-3}}$
Φ	the quantity minimised (Sect. 7)	–
$\lambda_F, \lambda_S, \lambda_Q$	weights of the terms of Φ	–
P_{ref}	prescribed mean pressure on the bottom plane	Pa

3. The equations of magnetohydrostatics

Three forces per unit volume act on a small volume of gas: the pressure force $-\nabla P$, gravity $-\rho g \hat{\mathbf{z}}$ and the Lorentz force $\mathbf{J} \times \mathbf{B}$ of the field on the currents in the ionised gas, with $\mathbf{J} = \nabla \times \mathbf{B}/\mu_0$ (Ampère's law). The gas is static if

$$\boxed{-\nabla P + \mathbf{J} \times \mathbf{B} - \rho g\, \hat{\mathbf{z}} = 0, \qquad \nabla \cdot \mathbf{B} = 0}$$

The second equation (no magnetic monopoles) holds for every real field. A vector identity splits the Lorentz force into

$$\mathbf{J} \times \mathbf{B} = -\nabla\left(\frac{B^2}{2\mu_0}\right) + \frac{(\mathbf{B} \cdot \nabla)\mathbf{B}}{\mu_0} :$$

a **magnetic pressure** $B^2/2\mu_0$ (field lines push apart) and a **magnetic tension** (curved field lines pull like stretched rubber bands). The ratio $\beta = 2\mu_0 P/B^2$ of gas to magnetic pressure decides which force dominates: deep down $\beta \gg 1$, high up $\beta \ll 1$. Where $\beta \ll 1$, a static state needs a nearly **force-free** field, $\mathbf{J} \times \mathbf{B} \approx 0$, with currents flowing along the field lines.

Without a field, the vertical equation is hydrostatic equilibrium $dP/dz = -\rho g$. With the ideal gas law $\rho = \bar{\mu} m_{\mathrm{H}} P/(k_B T)$ ($\bar{\mu}$ the mean particle mass in units of the hydrogen mass, which changes with ionisation; it comes from a tabulated equation of state) this becomes $d \ln P/dz = -1/H$ with the scale height $H = k_B T/(\bar{\mu} m_{\mathrm{H}} g)$, about 140 km in the photosphere.

4. Why the field has to change

Take the curl of the force balance. The curl of a gradient vanishes and $\nabla \times (\rho g \hat{\mathbf{z}}) = g(\partial_y \rho, -\partial_x \rho, 0)$, so a static state requires

$$\nabla \times (\mathbf{J} \times \mathbf{B}) = g\left(\frac{\partial \rho}{\partial y}, -\frac{\partial \rho}{\partial x}, 0\right).$$

Wherever hot and cool gas lie side by side at the same height (and in the Sun they do everywhere: hot rising granules next to cool sinking lanes), the density varies horizontally, and only the magnetic tension can hold that up. A field taken from a moving snapshot was never shaped for this, so with $\mathbf{B} = \mathbf{B}_{\rm obs}$ no pressure field can be static (the `mhs-poisson` modes show how far a pressure change alone gets). Here we let $\mathbf{B}$ change as well, *as little as possible*. That turns the problem into a compromise between "balance the forces" and "stay close to $\mathbf{B}_{\rm obs}$".

5. Making $\nabla \cdot \mathbf{B} = 0$ automatic: the vector potential

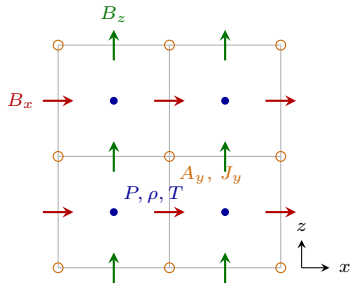
Instead of the field itself, the unknown is the vector potential $\mathbf{A}$, with $\mathbf{B} = \nabla \times \mathbf{A} + \bar{b}\,\hat{\mathbf{z}}$. The divergence of a curl is always zero, so $\nabla \cdot \mathbf{B} = 0$ holds automatically and never has to be imposed. $\bar{b}$ is needed because the box is periodic: the curl of a periodic potential has zero average vertical component over every horizontal plane, so the net vertical magnetic flux through the box has to be added separately.

6. The staggered grid

The quantities live at different places in each grid cell:

- P, ρ, T at the cell centre;
- each field component on the cell face it points through (B_x on the faces perpendicular to x, …); B_x there is the magnetic flux through the face divided by its area;
- $\mathbf{A}$ and the currents $\mathbf{J}$ on the cell edges.

The flux out of a cell is the sum over its six faces. For $\mathbf{B} = \nabla \times \mathbf{A}$ each face flux is a sum of $\mathbf{A}$ around the face's edges (Stokes' theorem), and every edge appears in two faces of the cell with opposite signs, so the total is *exactly* zero: $\nabla \cdot \mathbf{B} = 0$ holds on the grid itself, not just approximately.



An x–z slice. B_y, A_x, A_z, J_x, J_z lie on faces and edges not cut by this slice.

The field components and currents are differences of neighbouring values, e.g. $B_z = \partial A_y/\partial x - \partial A_x/\partial y + \bar{b}$ and $J_y = (\partial B_x/\partial z - \partial B_z/\partial x)/\mu_0$: a current is a *change* of the field across space, so a uniform field carries none. Each component of the force balance is evaluated on the faces of its own direction:

$$R_x = \frac{\partial P}{\partial x} - (J_y B_z - J_z B_y), \quad R_y = \frac{\partial P}{\partial y} - (J_z B_x - J_x B_z), \quad R_z = \frac{\partial P}{\partial z} + \rho g - (J_x B_y - J_y B_x).$$

Only the two *other* field components enter each one: a field along x exerts no force along x.

7. The quantity that is minimised

$$\Phi = \underbrace{\lambda_F \left\langle \frac{|\mathbf{R}|^2}{S^2} \right\rangle}_{\Phi_F} + \underbrace{\left\langle \frac{(\mathbf{B} - \mathbf{B}_{\rm obs})^2}{\sigma^2} \right\rangle}_{\Phi_B} + \underbrace{\lambda_S \left\langle \left| \frac{h^2\,\partial^2 (\mathbf{B} - \mathbf{B}_{\rm obs})}{\sigma} \right|^2 \right\rangle}_{\Phi_S} + \underbrace{\lambda_F \lambda_Q \left\langle \left| \frac{B_{\rm ref}\,\delta^2 \mathbf{J}}{S} \right|^2 \right\rangle}_{\Phi_Q}$$

$\langle \cdot \rangle$ is a volume-weighted average over the grid. Each term, and why it is there:

- Φ_F**, force balance.** The residual is divided by the local weight of the gas $S \approx \rho g$ (from the starting state, with a floor so it never becomes tiny), so $|\mathbf{R}|/S = 1$ means an unbalanced force as large as gravity. Without this scaling the deep, dense layers would dominate everything.
- Φ_B**, closeness to the given field.** Each face compares its own component with $\mathbf{B}_{\rm obs}$ interpolated to that face, in units of σ. If the deviations were random errors of size σ, Φ_B would be about 1 (this is the familiar χ^2 per data point). With a uniform σ of 5 G, any change of the field costs the same everywhere; there is deliberately no relative tolerance, because for observations the measurement noise of the Stokes profiles does not translate into a meaningful field error.
- Φ_S**, smoothness of the change.** The second differences ($h^2\partial^2$, with h the grid spacing) of $\mathbf{B} - \mathbf{B}_{\rm obs}$ are penalised: the model may differ from $\mathbf{B}_{\rm obs}$ by a smooth, large-scale correction, but not by cell-to-cell noise.
- Φ_Q**, no grid-scale currents.** Second differences of the currents along the grid are penalised. Without it the solver could balance forces with currents that flip sign from one cell to the next, which are numerical artefacts, not physics.

The weights λ set the trade-off. A larger λ_F favours force balance at the price of a larger change of the field; a smaller one keeps the field closer to $\mathbf{B}_{\rm obs}$ and leaves more force imbalance. There is no unique "right" value; the defaults are recorded with every run.

8. Boundary conditions and the pressure level

The box is periodic in x and y: what leaves one side enters the other. At the top and bottom the horizontal force equations are not used on the outermost layers (they lack the vertical derivative of the field there). The force balance fixes pressure *differences*; the overall pressure level is set by requiring the mean of $\ln P$ on the bottom plane to equal $\ln P_{\mathrm{ref}}$, a single number, not a pressure map. Two safeguards keep the pressure physical: in one step $\ln P$ may change by at most 5 anywhere, and P may never fall below e^{-3} times its starting value.

9. How the minimum is found

1. **Starting pressure.** Every column is put in exact hydrostatic equilibrium on the grid (field ignored), upward from P_{ref} at the bottom.
2. **Starting field.** With the pressure frozen, $\mathbf{A}$ and $\bar{b}$ are chosen so that $\nabla \times \mathbf{A} + \bar{b}\hat{\mathbf{z}}$ matches $\mathbf{B}_{\mathrm{obs}}$ as closely as possible ($\Phi_B + \Phi_S$ only). This is a linear least-squares problem.
3. **Joint steps.** The force residual depends non-linearly on the unknowns ($\mathbf{J} \times \mathbf{B}$ is quadratic in $\mathbf{B}$, ρ depends on P). Each step replaces it by its linear approximation in the changes of $\ln P$, $\mathbf{A}$ and $\bar{b}$ (Gauss–Newton, the least-squares form of Newton's method), solves for the change that minimises the approximated Φ, and accepts it only if the true Φ decreases. If it does not, the step is halved along the same direction (a backtracking line search) until it does.
4. **Stop** when Φ has stopped changing: its relative change is below 10^{-6} on three accepted steps in a row.

Many different potentials give the same field (adding a gradient to $\mathbf{A}$ does not change $\nabla \times \mathbf{A}$). Such changes have no effect on Φ; a tiny damping term keeps the linear systems well defined, and all results are judged by the physical quantities P and $\mathbf{B}$.

10. The visible surface

After every accepted step the optical depth at 500 nm, $\tau_{500}(z) = \tau_{\mathrm{top}} + \int_z^{z_{\mathrm{top}}} \chi_{500}\, dz'$, is recomputed in every column from the current pressure, and with it the height of $\tau_{500} = 1$, the visible surface. The opacity χ_{500} depends only on T and P, so the field changes the visible surface only through the pressure it demands. A strong flux tube is partly supported by its magnetic pressure, so its gas pressure is lower, its gas is more transparent, and its $\tau = 1$ surface lies deeper: the Wilson depression.

11. Summary

The mode looks for the static atmosphere closest to the snapshot. The temperature is kept exactly, $\nabla \cdot \mathbf{B} = 0$ holds exactly through the vector potential on a staggered grid, and the pressure and field are adjusted together to minimise a weighted sum of the force imbalance, the change of the field, and penalties against grid-scale noise. Unlike the pressure-only modes, it can in principle remove the buoyancy part of the imbalance, because only a changed field can supply the magnetic tension that holds up horizontal density differences.